

Rosenholz PM v2 — Klassendiagramm

Vollständiges Diagramm · alle Felder und Methoden · A3 Querformat

Kern-Entitäten (F16 / F22 / Person / Team / Dokument)

ProjectF16	+Double cpi	TaskF22	Person	Team	Document
	+Double spi	+String taskId	+String personId	+String teamId	+String documentId
+String projectId	+Double vac	+String taskCode	+String firstName	+String name	+String projectId
+String title	+String leadId	+String projectId	+String lastName	+String teamType	+String taskId
+String codename	+String teamId	+String parentTaskId	+String preferredName	+String leadId	+String authorId
+String phase	+String sponsorId	+String title	+String email	+String parentTeamId	+String approvedBy
+String status	+String milestones	+String description	+String phone	+Double budget	+String title
+String priority	+String scopeStatement	+String taskType	+String orgUnit	+String status	+String docCategory
+String complexity	+String createdAt	+String status	+String department	+String createdAt	+String docType
+String startDatePlanned	+String updatedAt	+String priority	+String location	+create(name, type) Team\$	+String version
+String endDatePlanned	+create(title, leadId) ProjectF16\$	+String assigneId	+String country	+loadById(id) Team\$	+String status
+String startDateActual	+loadById(id) ProjectF16\$	+String assignedBy	+String roleTitle	+addMember(personId, role, pct)	+String language
+String endDateActual	+loadRecent(n) List~ProjectF16~\$	+String milestones	+String employmentType		+String classification
+String currency		+String wbsCode	+Double dayRate	+save() bool	+String filePath
+Double budgetPlanned	+loadAll() List~ProjectF16~\$	+String startDatePlanned	+String status		+String fileUrl
+Double budgetActual	+recalcEV()	+String endDatePlanned	+String createdAt		+String fileHash
+Double earnedValue	+save() bool	+String startDateActual	+create(first, last, email, type) Person\$		+Int fileSize
+Double actualCost	+update() bool	+String endDateActual	+loadById(id) Person\$		+String summary
		+Int percentComplete	+search(fragment) List~Person~\$		+String createdAt
		+Double effortPlanned			+String updatedAt
		+Double effortActual	+save() bool		+create(title, type, projectId) Document\$
		+Double costActual			+loadById(id) Document\$
		+String workflowInstanceId			+loadForProject(pid) List~Document~\$
		+String workflowStatus			+loadRecent(n) List~Document~\$
		+String workflowCurrentState			+importLocalFile(path) bool
		+String createdAt			+snapshotVersion(note, by) bool
		+String updatedAt			+openFile(mode) bool
		+create(projectId, title) TaskF22\$			
		+loadById(id) TaskF22\$			+refreshFromUrl() bool
		+loadForProject(pid) List~TaskF22~\$			+save() bool
		+loadRecent(n) List~TaskF22~\$			
		+save() bool			
		+update() bool			

Repository & Archive (LMDB Dateispeicher)

DocumentRevision	ArchiveStore
+String documentId	«service singleton»
+uint32_t rev	
+uint32_t parentRev	+String stagingDir
+String revState	+init(basePath, mapBytes) bool\$
+Bool superseded	+stageContent(srcPath, outTmp) ChunkRef\$
+String contentHash	+commitContent(tmpPath, ref, docId, rev) bool\$
+Int64 contentSize	+retrieveContent(docId, rev, destPath) bool\$
+String createdBy	+chunkExists(sha256) bool\$
+String changeNote	+lookupRevChunk(docId, rev) String\$
+String createdAt	+deleteRevMapping(docId, rev) bool\$
+String updatedAt	
+transitionState(target) bool\$	+ensureFreeSpace() bool\$
+createRevision(docId, baseRev, by, note) DocumentRevision\$	
+loadByRev(docId, rev) DocumentRevision\$	
+loadAllRevisions(docId) List~DocumentRevision~\$	
+currentRevision(docId) DocumentRevision\$	
+latestRevNumber(docId) uint32_t\$	
+isTransitionAllowed(from, to) bool\$	
+save() bool	
+update() bool	
+recomputeSuperseded() bool	

Hinweis — DocumentRevision

5 states: *in_work* → *pre_released*, *locked*, *closed*

pre_released → *released*, *locked*, *closed*, *in_work*

released → *locked*, *closed* (*immutable*)

locked → *pre_released* (*newest rev only*), *closed*

closed → [*terminal*]

Exactly ONE revision per docId holds

superseded=false (the active revision).

Priority: *latest released* > *latest non-locked/non-closed* > *latest in_work*.

Flag swapped atomically in SQLite txn.

Hinweis — ArchiveStore

LMDB-backed, file content ONLY.

All metadata stays in SQLite.

Free-space guard: doubles map size when < 1 GB free (starts at 256 MB).

Content committed ONLY after successful state transition.

Namespace: Rosenholz::Archive

F18 Workflow-Hierarchie (Incidents / Risiken / Maßnahmen / ...)

F18Workflow	+String verifiedDate
	+String verifiedBy
+String vorgangId	+String phase
+String vorgangType	+String criteria
+String projectId	+String acceptanceCriteria
+String taskId	+String findings
+String parentVorgangId	+String gateResult
+String title	+String gateDecision
+String description	+String acType
+String status	+String validatedDate
+String ownerId	+String validatedBy
+String priority	+String impact
+String notes	+String audience
+String links	+String frequency
+String createdAt	+String channel
+String updatedAt	+String responsible
+String incidentType	+String lessonType
+String severity	+String recommendation
+String occurredDate	+String applicablePhases
+String resolvedDate	+String decisionType
+String rootCause	+String rationale
+String immediateAction	+String decisionDate
+String resolution	+String decisionBy
+Double costImpact	+String alternativesConsidered
+Int scheduleImpactDays	+String changeType
+String scopeImpact	+String justification
+String qualityImpact	+String crImpact
+String riskLevel	+String raisedDate
+Int probabilityScore	+String crDecisionDate
+Int impactScoreTime	+String crDecisionRationale
+Int impactScoreCost	+Int crScheduleImpactDays
+Int impactScoreQuality	+String executedBy
+Int impactScoreScope	+String executionDate
+Int overallRiskScore	+create(projectId, title, type) F18Workflow\$
+String responseStrategy	+loadById(id) F18Workflow\$
+String contingencyPlan	+loadForProject(pid, type) List~F18Workflow~\$
+String triggerCondition	+loadForTask(tid, type) List~F18Workflow~\$
+String residualRiskLevel	+loadRecent(n) List~F18Workflow~\$
+Double costReserve	+addStep(title, stepType) F18WorkflowStep
+Int scheduleReserveDays	+addNote(authorId, text) bool
+String measureCategory	
+String plannedDate	+loadSteps()
+String actualDate	+recalcRiskScore()
+String effectiveness	+save() bool
+String verificationMethod	+update() bool

F18WorkflowStep	+Bool requiresDocument
	+String decision
+String stepId	+String decisionBy
+String vorgangId	+String decisionDate
+String tplStepId	+String comment
+String title	+String trackingStatus
+String description	+String plannedDate
+String stepType	+String focusDate
+Int sequenceOrder	+String archivedDate
+String executionType	+String priority
+String predecessorStepIds	+String assignedToGroup
+Bool isInitialize	+String progressNote
+Bool isFinal	+Int percentComplete
+String assignedTo	+String notes
+String requiredRole	+String createdAt
+String dueDate	+String updatedAt
+String startedDate	+canStart(allSteps) bool
+String completedDate	+loadById(id) F18WorkflowStep\$
+Int slaHours	+loadForVorgang(vorgangId) List~F18WorkflowStep~\$
+Bool slaBreached	
+String status	+isComplete() bool
+Bool autoApprove	+save() bool
+Bool requiresComment	+remove() bool

Communication
+String commId
+String ownerId
+String ownerType
+String commType
+String title
+String agenda
+String scheduledDate
+String actualDate
+Int durationMins
+String channel
+String location
+String organiserId
+String participants
+String decisions
+String actions
+String notes
+String status
+String createdAt
+String updatedAt
+create(ownerId, ownerType, title, type) Communication\$
+loadById(id) Communication\$
+loadForOwner(id, ownerType) List~Communication~\$
+complete(decisions, actions) bool
+loadRecent(n) List~Communication~\$
+save() bool
+update() bool
+remove() bool

vorgangType ∈ generic incident risk measure qualityGate assumptionConstraint communicationPlan lessonsLearned decisionLog changeRequest changeObject
Kardinalitäten pro F22: assumptionConstraint : 0..1 lessonsLearned : 0..1 decisionLog : 0..1 alle anderen : 0..*
ChangeObject: parentVorgangId → ChangeRequest

Hinweis — F18WorkflowStep stepType ∈ task approval review notification executionType ∈ sequential parallel free status ∈ pending in_progress approved rejected skipped trackingStatus ∈ planned focused due archived
Jeder F18Workflow hat genau: 1× Init-Step (isInitialize=true, autoApprove=true) N× Mid-Steps 1× End-Step (isFinal=true)
End-Step wird nie als Predecessor erlaubt.

Hinweis — Communication ownerType ∈ project task f18step commType ∈ message call meeting email report status ∈ scheduled completed cancelled
Ersetzt Meeting vollständig.

Workflow Engine (WFI / WFA / Template)

WorkflowTemplate
+String templateId
+String name
+String version
+String description
+String executionType
+String entityTypees
+Bool slaEnforced
+Int defaultSlaHours
+String createdAt
+create(name, execType) WorkflowTemplate\$
+loadById(id) WorkflowTemplate\$
+loadForEntityType(type) List~WorkflowTemplate~\$
+loadAll() List~WorkflowTemplate~\$
+save() bool

WorkflowInstance
+String instanceId
+String templateId
+String name
+String entityType
+String entityId
+String executionType
+String status
+String initiatedBy
+String initiatedDate
+String dueDate
+String completedDate
+String slaBreachDate
+Bool slaBreached
+String escalatedTo
+String escalatedDate
+String priority
+String outcome
+String notes
+String createdAt
+String updatedAt
+create(type, id, name, exec) WorkflowInstance\$
+loadById(id) WorkflowInstance\$
+loadForEntity(type, id) List~WorkflowInstance~\$
+loadActive() List~WorkflowInstance~\$
+loadBreached() List~WorkflowInstance~\$
+isComplete() bool
+save() bool

WorkflowAction	+String status
	+Bool autoApprove
+String actionId	+Bool requiresComment
+String instanceId	+Bool requiresDocument
+String tplActionId	+String decision
+String title	+String decisionBy
+String description	+String decisionDate
+Int sequenceOrder	+String comment
+String executionType	+String trackingStatus
+String predecessorActionIds	+String priority
+Bool isInitialize	+String progressNote
+Bool isFinal	+Int percentComplete
+String assignedTo	+String notes
+String requiredRole	+String createdAt
+String dueDate	+String updatedAt
+String startDate	+canStart(allActions) bool
+String completedDate	
+Int slaHours	+isComplete() bool
+Bool slaBreached	+save() bool

WorkflowEngine
« <i>service</i> »
+startFromTemplate(templateId, entityType, entityId, name) WorkflowInstance\$
+startAdHoc(entityType, entityId, name, execType) WorkflowInstance\$
+addAction(inst, title, execType, seq, preds) WorkflowAction\$
+tick(inst) bool\$
+fireAction(inst, actionId, decision, actorId) bool\$
+escalate(inst, personId, reason) bool\$
+searchInstances(entityType, status, nameContains, slaOnly) List~WorkflowInstance~\$
+attachDocumentToInstance(instanceId, documentId, rel) bool\$
+attachDocumentToAction(actionId, documentId, rel) bool\$
+loadDocumentsForInstance(instanceId) List~Document~\$
+loadDocumentsForAction(actionId) List~Document~\$
+createDecisionLogEntry(actionId, entityId, title, rationale) bool\$
+syncEntityWorkflowFields(inst) bool\$
+runAll() bool\$

Hinweis — WorkflowEngine

Klassische WFI/WFA-Engine für Prozess-
Workflows (Genehmigungen, Eskalationen).

F18WorkflowStep ist das Äquivalent
für F18-Vorgänge.

System / Migration

MigrationEngine
« <i>service</i> »
+runFor(dbName) bool\$
+currentVersion(dbName) int\$
+targetVersion(dbName) int\$
+applyMigration(db, migration) bool\$
+runAll() bool\$
+registry() List~Migration~\$
+targetVersionMap() Map\$

SchemaVersions
« <i>constants</i> »
+core = 2\$
+projects = 2\$
+workflow = 2\$
+documents = 2\$
+tracking = 2\$
+f18 = 2\$

Hinweis — MigrationEngine

v2 Baseline: registry() ist leer.

Frische DBs starten direkt bei v2.

Keine Rückwärtskompatibilität.

Zukünftige Deltas kommen in registry().

Beziehungen (vollständig)

note for F18Workflow "vorgangType ∈ generic incident risk measure\n qualityGate assumptionConstraint\n communicationPlan lessonsLearned\n decisionLog changeRequest changeObject\n\nKardinalitäten pro F22:\nassumptionConstraint : 0..1\nlessonsLearned : 0..1\ndecisionLog : 0..1\nalle anderen : 0..*\n\nChangeObject: parentVorgangId → ChangeRequest" ProjectF16 "1" --> "0..1" Person : lead Team "0..1" --> "0..1" Person : lead F18Workflow "1" *-- "2..*" F18WorkflowStep : Init + Mid + End\nKomposition TaskF22 "1" --> "0..*" Communication : ownerType=task TaskF22 "1" --> "0..*" Document : taskId WorkflowEngine ..> WorkflowTemplate : verwaltet WorkflowInstance --> Document : entity_documents (Verlinkung) WorkflowInstance --> TaskF22 : entityType=task DocumentRevision ..> ArchiveStore : commitContent / retrieveContent	ProjectF16 "1" --> "0..*" TaskF22 : enthält ProjectF16 "1" --> "0..1" Team : team ProjectF16 "1" --> "0..*" F18Workflow : besitzt (projectId) F18Workflow "1" --> "0..1" F18Workflow : parentVorgangId\n(nur changeObject → changeRequest) F18WorkflowStep "1" --> "0..*" Communication : ownerType=f18step WorkflowTemplate "1" --> "0..*" WorkflowInstance : instanziiert WorkflowEngine ..> WorkflowInstance : steuert WorkflowAction --> Document : entity_documents (Verlinkung) WorkflowInstance --> Document : entityType=document WorkflowEngine ..> DocumentRevision : transitionState auf End-Schritt	TaskF22 "0..1" --> "0..*" TaskF22 : parentTaskId (Teilaufgaben) TaskF22 "0..1" --> "0..1" Person : assignee TaskF22 "1" --> "0..*" F18Workflow : besitzt (taskId) ProjectF16 "1" --> "0..*" Communication : ownerType=project ProjectF16 "1" --> "0..*" Document : projectId WorkflowInstance "1" *-- "2..*" WorkflowAction : Komposition WorkflowEngine ..> WorkflowAction : feuert / tickt WorkflowInstance --> ProjectF16 : entityType=project Document "1" --> "0..*" DocumentRevision : docId:rev Revisionen MigrationEngine ..> SchemaVersions : liest Zielversionen
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Rosenholz PM v2 — Vereinfachtes Klassendiagramm

Übersicht · max. 6 Schlüsselfelder pro Klasse · Methodennamen · Beziehungsstruktur

- Kern-Entitäten (F16 / F22 / Person / Team / Dokument)
- F18 Workflow (Incident / Risk / Measure / ...)
- Workflow Engine (WFI / WFA / Template)
- Repository / Archive (LMDB Inhaltsspeicher + Dokument-Revisionen)
- System (Migration / Schema-Versionen)

ProjectF16	TaskF22	Person	Team	Document
+String projectId	+String taskId	+String personId	+String teamId	+String documentId
+String title	+String taskCode	+String firstName	+String name	+String projectId
+String codename	+String projectId	+String lastName	+String teamType	+String taskId
+String phase	+String parentTaskId	+String preferredName	+String leadId	+String authorId
+String status	+String title	+String email	+String parentTeamId	+String approvedBy
+String priority	+String description	+String phone	+Double budget	+String title
... +23 Felder	... +24 Felder	... +12 Felder	... +5 Felder	... +20 Felder
+loadAll() List~ProjectF16~\$	+save() bool	+save() bool	+save() bool	+refreshFromUrl() bool
+recalcEV()	+update() bool			+save() bool
+save() bool				
+update() bool				
... +1 Methoden				

DocumentRevision	ArchiveStore	F18Workflow	F18WorkflowStep	Communication
+String documentId	«service singleton»	+String vorgangId	+String stepId	+String commId
+uint32_t rev	+String stagingDir	+String vorgangType	+String vorgangId	+String ownerId
+uint32_t parentRev	+init(basePath, mapBytes) bool\$	+String projectId	+String tplStepId	+String ownerType
+String revState	+stageContent(srcPath, outTmp) ChunkRef\$	+String taskId	+String title	+String commType
+Bool superseded	+commitContent(tmpPath, ref, docId, rev) bool\$	+String parentVorgangId	+String description	+String title
+String contentHash	+retrieveContent(docId, rev, destPath) bool\$	+String title	+String stepType	+String agenda
... +12 Felder	+chunkExists(sha256) bool\$	... +77 Felder	... +34 Felder	... +18 Felder
+save() bool	... +2 Felder	+loadSteps()	+isComplete() bool	+save() bool
+update() bool	+ensureFreeSpace() bool\$	+recalcRiskScore()	+save() bool	+update() bool
+recomputeSuperseded() bool		+save() bool	+remove() bool	+remove() bool
		+update() bool		
		... +1 Methoden		

WorkflowTemplate	WorkflowInstance	WorkflowAction	WorkflowEngine	MigrationEngine
+String templateId	+String instanceId	+String actionId	« <i>service</i> »	« <i>service</i> »
+String name	+String templateId	+String instanceId	+startFromTemplate(templateId, entityType, entityId, name) WorkflowInstance\$	+runFor(dbName) bool\$
+String version	+String name	+String tplActionId	+startAdHoc(entityType, entityId, name, execType) WorkflowInstance\$	+currentVersion(dbName) int\$
+String description	+String entityType	+String title	+addAction(inst, title, execType, seq, preds) WorkflowAction\$	+targetVersion(dbName) int\$
+String executionType	+String entityId	+String description	+tick(inst) bool\$	+applyMigration(db, migration) bool\$
+String entityTypes	+String executionType	+Int sequenceOrder	+fireAction(inst, actionId, decision, actorId) bool\$	+runAll() bool\$
... +6 Felder	... +17 Felder	... +27 Felder	+escalate(inst, personId, reason) bool\$	+registry() List~Migration~\$
+loadAll() List~WorkflowTemplate~\$	+loadActive() List~WorkflowInstance~\$	+isComplete() bool	... +7 Felder	+targetVersionMap() Map\$
+save() bool	+loadBreachd() List~WorkflowInstance~\$	+save() bool	+runAll() bool\$	
	+isComplete() bool			
	+save() bool			

SchemaVersions
« <i>constants</i> »
+core = 2\$
+projects = 2\$
+workflow = 2\$
+documents = 2\$
+tracking = 2\$
+f18 = 2\$

Kernbeziehungen

ProjectF16 "1" --> "0..*" TaskF22 : enthält	TaskF22 "0..1" --> "0..*" TaskF22 : parentTaskId (Teilaufgaben)	ProjectF16 "1" --> "0..1" Person : lead
ProjectF16 "1" --> "0..1" Team : team	TaskF22 "0..1" --> "0..1" Person : assignee	Team "0..1" --> "0..1" Person : lead
ProjectF16 "1" --> "0..*" F18Workflow : besitzt (projectId)	TaskF22 "1" --> "0..*" F18Workflow : besitzt (taskId)	F18Workflow "1" --> "0..1" F18Workflow : parentVorgangId\n(nur changeObject → changeRequest)
		F18WorkflowStep "1" --> "0..*" Communication : ownerType=f18step
ProjectF16 "1" --> "0..*" Communication : ownerType=project	TaskF22 "1" --> "0..*" Communication : ownerType=task	WorkflowTemplate "1" --> "0..*" WorkflowInstance : instanziiert
ProjectF16 "1" --> "0..*" Document : projectId	TaskF22 "1" --> "0..*" Document : taskId	WorkflowInstance --> ProjectF16 : entityType=project
WorkflowInstance --> Document : entity_documents (Verlinkung)	WorkflowAction --> Document : entity_documents (Verlinkung)	Document "1" --> "0..*" DocumentRevision : docId:rev Revisionen
WorkflowInstance --> TaskF22 : entityType=task	WorkflowInstance --> Document : entityType=document	

Design-Hinweise

Hinweis — F18Workflow
vorgangType ∈ generic incident risk measure qualityGate assumptionConstraint communicationPlan lessonsLearned decisionLog changeRequest changeObject
Kardinalitäten pro F22: assumptionConstraint : 0..1 lessonsLearned : 0..1 decisionLog : 0..1 alle anderen : 0..*
ChangeObject: parentVorgangId → ChangeRequest

Hinweis — F18WorkflowStep

stepType ∈ *task* | *approval* | *review* | *notification*
executionType ∈ *sequential* | *parallel* | *free*
status ∈ *pending* | *in_progress* | *approved* | *rejected* | *skipped*
trackingStatus ∈ *planned* | *focused* | *due* | *archived*

Jeder *F18Workflow* hat genau:
1× *Init-Step* (*isInitialize=true*, *autoApprove=true*)
N× *Mid-Steps*
1× *End-Step* (*isFinal=true*)

End-Step wird nie als *Predecessor* erlaubt.

Hinweis — Communication

ownerType ∈ *project* | *task* | *f18step*
commType ∈ *message* | *call* | *meeting* | *email* | *report*
status ∈ *scheduled* | *completed* | *cancelled*

Ersetzt *Meeting* vollständig.

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LMDB-backed, *file content ONLY*.
All metadata stays in SQLite.
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when < 1 GB free (starts at 256 MB).
Content committed ONLY after
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Namespace: Rosenholz::Archive

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5 states: *in_work* → *pre_released*, *locked*, *closed*
pre_released → *released*, *locked*, *closed*, *in_work*
released → *locked*, *closed* (*immutable*)
locked → *pre_released* (*newest rev only*), *closed*
closed → [*terminal*]

Exactly *ONE* revision per *docId* holds
superseded=false (*the active revision*).
Priority: latest released > latest
non-locked/non-closed > latest in_work.
Flag swapped atomically in SQLite txn.